

# Youniss Kandah

Applied AI & Backend Engineer | Python, FastAPI, React, Retrieval, Document AI, Production Systems  
Austria-based | Open to relocate

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Building production AI services, retrieval and document-processing pipelines, and safety-critical backend systems. Currently contributing to the Austrian FFG-funded AI4Emergency project at ONDEWO.

## EXPERIENCE

### AI Engineer, ONDEWO ([ondewo.com](https://www.ondewo.com)), Remote, Austria

Mar 2026 – Present

- Contributing to AI4Emergency, an Austrian funded research project developing AI-driven communication technologies for safety-critical emergency operations in disaster and large-event scenarios.
- Built a web-crawling and ingestion pipeline for retrieval-augmented generation and exposed it through a gRPC facade for downstream integration.
- Created an automated LLM QA pipeline that tests agent changes for regressions and enables TDD for agents.
- Created production-grade MCP servers and LangGraph agents for emergency-service workflows.

### Software Engineer, ib-data ([abk.at](https://www.abk.at)), Vienna, Austria

Oct 2022 – Feb 2026

- Built AI features including semantic search with PostgreSQL/pgvector + Python and a FastAPI pipeline that converted PDFs into structured data using LLM/VLM processing with human review.
- Built a customer self-service portal ([kundenportal.abk.at](https://kundenportal.abk.at)) using React, Spring Boot, MS SQL, JWT authentication, and Dockerized Ubuntu deployment.
- Migrated a legacy construction platform ([abk9.com](https://www.abk9.com)) to a modern React application and established CI/CD (TeamCity), automated tests, and Linux VM deployment workflows in a Scrum team.
- Replaced a paid documentation tool with an open-source stack (TinaCMS), removing its licence cost.
- Trained a YOLOv8 model, with labelling pipelines in Label Studio, to detect doors, windows and stairs in construction drawings.
- Prototyped a document-to-specification pipeline converting PDF and Excel sources into standardised Austrian construction descriptions.
- Delivered a semantic search proof of concept on pgvector that improved both relevance and latency over the legacy full-text search.
- Built a LangChain + FastAPI microservice summarising construction documents for a legacy Delphi backend, deployed on-premise with Docker and TeamCity.
- Built a large enterprise UI with React, Kendo, Zustand, React Query, Storybook and Playwright for a multi-year rollout.
- Automated recurring processes on Linux VMs, cutting manual operational work.

### Freelance AI Engineer (part-time), Austria

Dec 2024 – Feb 2026

- Built and deployed a LangChain-based automated grading service integrated into an LMS, optimizing for performance and memory efficiency.
- Taught multiple AI-related programs for different coding schools (LLMs, multi-modal, frontend development).

## EDUCATION

### MSc in Artificial Intelligence

JKU, Linz, Austria

Feb 2026 – paused

Selected coursework: AI in Life Sciences, Biochemistry, Bioanalytics, Medical technology.

### BSc in Artificial Intelligence

JKU, Linz, Austria

Oct 2023 – Feb 2026

Thesis: Deep learning for language-based audio retrieval.

## DOMAIN EXPERIENCES

### Health care: Samariterbund, Paramedic (9 months)

Direct exposure to emergency-response workflows, patient-facing care, and time-critical operational environments.

### Construction: HTL Mödling, Student (5 years); ABK, Software Engineer (3 years)

Training in construction engineering and building production software for construction management.

## RECOGNITION & PROJECTS

- DCASE 2025:** Competed in the DCASE 2025 challenge on language-based audio retrieval and published a peer-reviewed technical report ([results](#), [technical report](#)).
- On-device LLM:** Trained a small language model from scratch and shipped it as an on-device iOS app — a self-directed project taken end to end ([project](#), [App Store](#)).
- Publication:** Survey paper on LLM-based recommender systems ([paper](#)).
- Certification:** NVIDIA Certified Professional: Agentic AI ([credential](#)).
- Degree pace:** Completed the BSc ahead of the standard program duration while working full-time/part-time alongside studies.

## SKILLS

**Languages:** Python, TypeScript/JavaScript, Java, SQL.

**Applied AI:** Retrieval/RAG, LLM/VLM document extraction, evaluation, guardrails, inference serving.

**Backend:** FastAPI, Spring Boot, REST APIs, gRPC, PostgreSQL/pgvector.

**Frontend:** React, Next.js, Tailwind.

**Infra / Ops:** Docker, AWS, Azure, Linux, CI/CD, Kubernetes, Airflow.

**Tools:** LangChain, DSPy, NeMo, NIM, vLLM, PyTorch.  
**Spoken:** English (Fluent), German (Native), Arabic (A2).